

$$u(x, t)$$

$$u_{tt} = a^2 u_{xx}$$

$$u(x, t) = f(x+at) + g(x-at)$$

$$x+at = C_1$$

$$x-at = C_2$$

$$\begin{cases} \xi = x+at \\ \eta = x-at \end{cases}$$

$$u_{\xi\eta} = 0$$

$$f(\xi) + g(\eta) \\ \forall C^2 \quad \forall C^2$$

$$\text{I } \forall f(\xi) \in C^2, \forall g(\eta) \in C^2$$

$$f(x+at) + g(x-at) \text{ пер. ур.}$$

$$\text{II } \tilde{u}(x, t) \text{ пер. ур.}$$

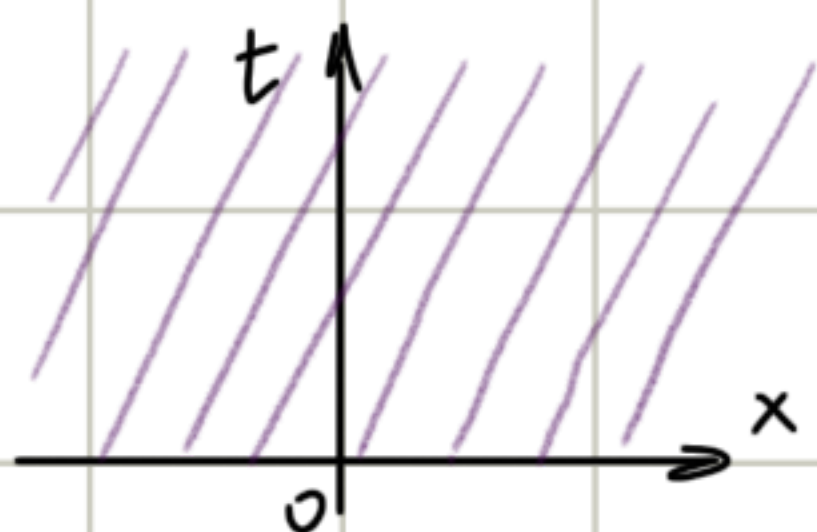
$$\exists \tilde{f}(\xi) \quad \exists \tilde{g}(\eta)$$

$$\tilde{u}(x, t) = \tilde{f}(x+at) + \tilde{g}(x-at)$$

Лемма Коши для волнового ур-я (функции $\tilde{f}$ и $\tilde{g}$ непрерывны)

$$u(x, t)$$

$$(1) u_{tt} - a^2 u_{xx} = F(x, t), \quad x \in \mathbb{R}^1, t > 0$$



$$(2) u|_{t=0} = \varphi(x), \quad x \in \mathbb{R}^1$$

$$(3) u_t|_{t=0} = \psi(x), \quad x \in \mathbb{R}^1$$

Опр.

Φ -я $u(x, t)$ наз. пер. (1), (2), (3), если

$$1) u(x, t) \in C^2 (x \in \mathbb{R}^1, t \geq 0)$$

$$2) \text{ угодн. все (1), (2), (3)}$$

Итак утвердим, что $u(x, t)$ - пер.

$$\Downarrow \\ \varphi(x) \in C^2(\mathbb{R}^1), \psi(x) \in C^1(\mathbb{R}^1)$$

$$F(x, t) \in C(x \in \mathbb{R}^1, t > 0)$$

$$\nexists F(x, t) \equiv 0$$

$$(1_0) u_{tt} - a^2 u_{xx} = 0, \quad x \in \mathbb{R}^1, t > 0$$

$$(2), (3)$$

$$u(x, t) = f(x+at) + g(x-at)$$

$$u_t(x, t) = f'(x+at) \cdot a + g'(x-at) \cdot (-a)$$

↑ Неотх. ур. $\exists$ -я пер.

$$u|_{t=0} = f(x) + g(x) = \psi(x), x \in \mathbb{R}^1$$

$$u_t|_{t=0} = a f'(x) - a g'(x) = \psi(x), x \in \mathbb{R}^1$$

$$\begin{cases} f'(x) + g'(x) = \psi'(x), x \in \mathbb{R}^1 \\ f'(x) - g'(x) = \frac{1}{a} \psi(x), x \in \mathbb{R}^1 \end{cases}$$

$$f'(x) = \frac{1}{2} \psi'(x) + \frac{1}{2a} \psi(x), x \in \mathbb{R}^1 \Rightarrow f(x) = \frac{1}{2} \psi(x) + \frac{1}{2a} \int_0^x \psi(\eta) d\eta + A, x \in \mathbb{R}^1$$

$$g(x) = \frac{1}{2} \psi(x) - \frac{1}{2a} \int_0^x \psi(\eta) d\eta - A, x \in \mathbb{R}^1$$

$$u(x, t) = \left[\frac{1}{2} \psi(x+at) + \frac{1}{2a} \int_0^{x+at} \psi(\eta) d\eta + A \right] + \left[\frac{1}{2} \psi(x-at) - \frac{1}{2a} \int_0^{x-at} \psi(\eta) d\eta - A \right]$$

$\forall (x, t) \in \mathbb{R}^2$ $(x+at) - t$ $(x-at) - t$

$$\begin{matrix} x+at & t \\ x-at & t \end{matrix} \Leftrightarrow \forall (x, t) \in \mathbb{R}^2$$

$$u(x, t) = \frac{\psi(x+at) + \psi(x-at)}{2} + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\eta) d\eta \quad \forall (x, t) \in \mathbb{R}^2$$

(4)

ф-на
Даламбера

$$\exists \psi(x) \in C^2(\mathbb{R}^1)$$

$$\psi(x) \in C^1(\mathbb{R}^1)$$

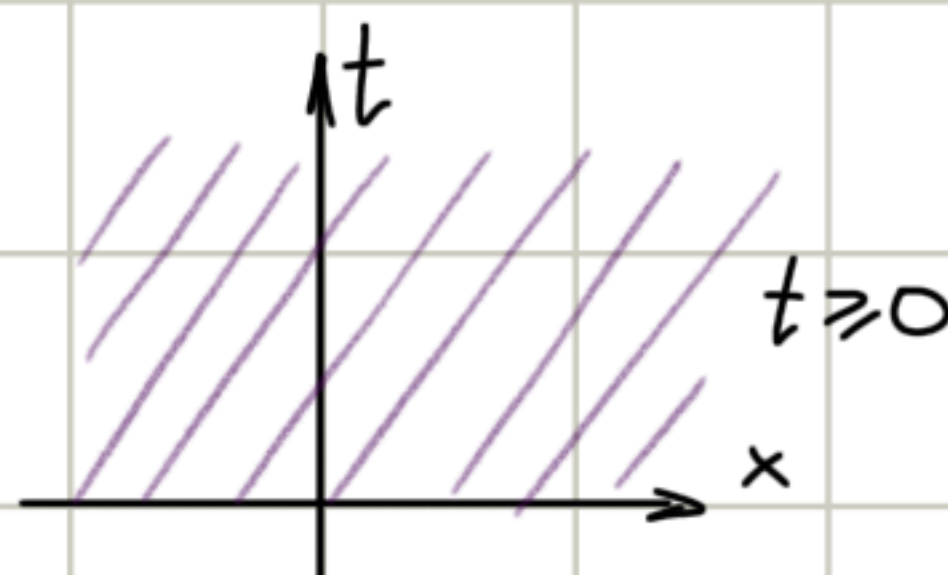
$\Rightarrow u(x, t)$ опрег. (4) есмб ф-на

$$C^2(\mathbb{R}^2)$$

$$y.g. (1)$$

$$y.g. (2), y.g. (3)$$

$$\psi(\xi), \xi = x+at$$



$$a \quad b$$

$$h(t) \in C[a, b]$$

$$H(x) = \int_a^x h(t) dt$$

$$\Rightarrow H(x) \in C^1[a, b]$$

$$H'(x) = h(x)$$

th

$$\exists \psi(x) \in C^2(\mathbb{R}^1), \psi(x) \in C^1(\mathbb{R}^1), \text{ тогда задача}$$

$$(1) u_{tt} - a^2 u_{xx} = 0, x \in \mathbb{R}^1, t > 0$$

$$(2) u|_{t=0} = \psi(x), x \in \mathbb{R}^1$$

$$(3) u_t|_{t=0} = \psi(x), x \in \mathbb{R}^1$$

имеет решение (4), причем это реш. уравн.

$$1) u_1(x, t) - \text{реш.} \begin{cases} u_{tt} - a^2 u_{xx} = F(x, t) \\ u|_{t=0} = \varphi_1(x), x \in \mathbb{R}^1 \\ u_t|_{t=0} = \psi_1(x), x \in \mathbb{R}^1 \end{cases}$$

$$2) u_2(x, t) - \text{реш.} \begin{cases} u_{tt} - a^2 u_{xx} = F(x, t) \\ u|_{t=0} = \varphi_2(x), x \in \mathbb{R}^1 \\ u_t|_{t=0} = \psi_2(x), x \in \mathbb{R}^1 \end{cases}$$

$$3) F(x, t), \varphi_1(x), \varphi_2(x), \psi_1(x), \psi_2(x) \text{ y.g. HY}$$

$$v(x, t) = u_1(x, t) - u_2(x, t) = \frac{1}{2} (\Phi(x+at) + \Phi(x-at)) + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\eta) d\eta$$

$$\begin{cases} v_{tt} - a^2 v_{xx} = 0, & x \in \mathbb{R}', t > 0 \\ v|_{t=0} = \varphi_1(x) - \varphi_2(x) = \Phi(x) \in C^2(\mathbb{R}') \\ v_t|_{t=0} = \psi_1(x) - \psi_2(x) = \Psi(x) \in C^1(\mathbb{R}') \end{cases}$$

$$|v(x, t)| \leq \underbrace{\frac{1}{2} (|\Phi(x+at)| + |\Phi(x-at)|)}_{< \delta} + \underbrace{\frac{1}{2a} \int_{x-at}^{x+at} |\Psi(\eta)| d\eta}_{< \delta} \leq \frac{1}{2a} \delta \int_{x-at}^{x+at} 1 dx = \frac{1}{2a} \delta 2at = \delta t \leq \delta T$$

$$1) 2) 3) \Rightarrow \forall \varepsilon > 0 \forall T > 0 \exists \delta = \delta(\varepsilon, T) > 0$$

$$\delta \in (0, \frac{\varepsilon}{1+T}]$$

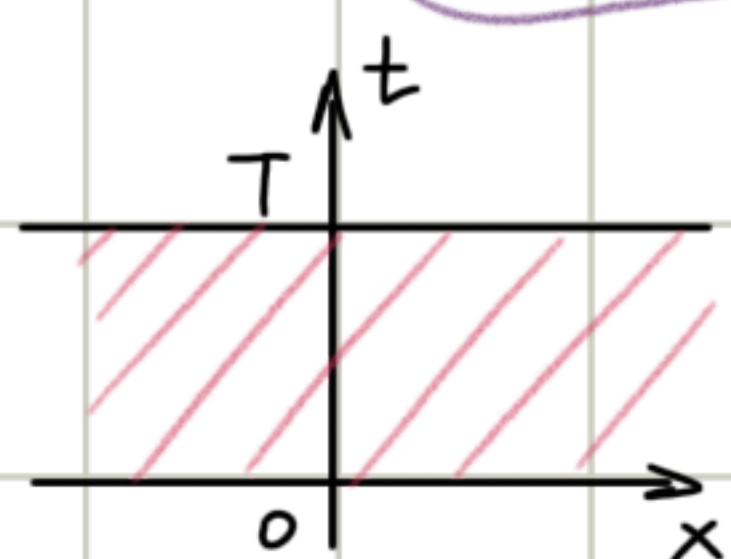
$$\text{тогда } |\varphi_1(x) - \varphi_2(x)| < \delta, \forall x \in \mathbb{R}'$$

$$(|\Phi(x)| < \delta, \forall x \in \mathbb{R}')$$

$$|\psi_1(x) - \psi_2(x)| < \delta, \forall x \in \mathbb{R}'$$

$$(|\Psi(x)| < \delta, \forall x \in \mathbb{R}')$$

$$\text{то } |u_1(x, t) - u_2(x, t)| < \varepsilon \quad \forall x \in \mathbb{R}', \forall t: 0 \leq t \leq T$$



$$|v(x, t)| \leq \delta + \delta T = \delta(1+T)$$

$$\varepsilon = \delta(1+T)$$

$$|v(x, t)| \leq \varepsilon$$

$$\delta = \frac{\varepsilon}{1+T}$$

$a=1$ Локализированная задача Коши

$$(5) \quad u_{tt} - u_{xx} = 0, \quad x \in \mathbb{R}', t > 0$$

$$(6) \quad u|_{t=0} = \varphi(x) \in C^2, \quad 0 < x < 1$$

$$(7) \quad u_t|_{t=0} = \psi(x) \in C^1, \quad 0 < x < 1$$

$$u(x, t) = f(x+t) + g(x-t)$$

$$\begin{cases} u|_{t=0} = f(x) + g(x) = \varphi(x), & 0 < x < 1 \end{cases}$$

$$\begin{cases} u_t|_{t=0} = f'(x) - g'(x) = \psi(x), & 0 < x < 1 \end{cases}$$

$$+ \begin{cases} f'(x) + g'(x) = \psi'(x), & 0 < x < 1 \end{cases}$$

$$\begin{cases} f'(x) - g'(x) = \psi(x), & 0 < x < 1 \end{cases}$$

$$f'(x) = \frac{1}{2} \psi'(x) + \frac{1}{2} \psi(x), \quad 0 < x < 1$$

$$f(x) = \frac{1}{2} \varphi(x) + \frac{1}{2} \int_0^x \psi(\eta) d\eta + A, \quad 0 < x < 1$$

$$g(x) = \frac{1}{2} \varphi(x) - \frac{1}{2} \int_0^x \psi(\eta) d\eta - A, \quad 0 < x < 1$$

$$u(x, t) = \left[f(x+t) \right] + \left[g(x-t) \right] = \left[\frac{1}{2} \varphi(x+t) + \frac{1}{2} \int_0^{x+t} \psi(\eta) d\eta + A \right] + \left[\frac{1}{2} \varphi(x-t) - \frac{1}{2} \int_0^{x-t} \psi(\eta) d\eta - A \right]$$

$$(x, t): 0 < x + t < 1$$

$$(x, t) : 0 < x - t < 1$$

$$b \quad \begin{cases} 0 < x+t < 1 \\ 0 < x-t < 1 \end{cases}$$

$$0 < x + t < 1$$

$$0 = x_4 t$$

$$x + t = 1$$

$$-x = t$$

$$t = 1 - x$$

$$x = t$$

$$x-1=t$$

